

Bluestock Mutual Fund Analytics Capstone

End-to-End Data Engineering, Dashboarding & Risk Analysis

Submitted To: Bluestock Fintech

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Table of Contents

- 1. Executive Summary
- 2. Data Architecture & Ingestion
- 3. ETL Process & Data Cleaning
- 4. Exploratory Data Analysis (EDA)
- 5. Fund Performance Analytics
- 6. Advanced Risk Metrics & Modeling
- 7. Dashboard Implementation
- 8. Fund Recommendation Engine
- 9. Strategic Recommendations
- 10. Limitations & Future Scope

1. Executive Summary

This report details the end-to-end development of the Bluestock Mutual Fund Analytics platform, a comprehensive 7-day capstone project. The primary objective was to architect a robust data pipeline capable of ingesting, cleaning, and analyzing massive volumes of mutual fund data, ultimately surfacing actionable insights through interactive dashboards and recommendation algorithms.

Over the course of the project, 17 raw datasets comprising over 127,000 rows of data were ingested from AMFI public sources. A local SQLite database (bluestock_mf.db) was engineered using a Star Schema design to act as the single source of truth. Leveraging Python (Pandas, NumPy, SciPy), advanced financial metrics - including Sharpe Ratios, Value at Risk (VaR), and Alpha/Beta - were calculated for 40 mutual fund schemes.

Finally, an interactive 4-page Power BI dashboard was deployed, accompanied by a Python-based algorithmic fund recommendation engine capable of suggesting optimized funds based on investor risk profiles.

Key Achievements

- **Data Pipeline:** Automated ETL processing 127,000+ rows across 17 datasets.
- **Database:** Star-schema SQLite DB with 11 validated tables and 100% row-count match.
- **Analytics:** Calculated CAGR, Sharpe, Sortino, Alpha, Beta, VaR, CVaR, and HHI for all funds.
- **Dashboard:** 4-page interactive Power BI dashboard with drill-through and slicers.
- **Recommender:** Algorithmic engine mapping risk profiles to Top 3 funds by Sharpe Ratio.

2. Data Architecture & Ingestion

The foundation of the analytics platform relies on diverse data sources, simulating real-world API pulls and batch file drops from financial institutions. A total of 10 core CSV datasets were sourced from the Association of Mutual Funds in India (AMFI) and supplementary financial data providers.

2.1 Data Sources

- **Fund Master (01_fund_master.csv):** Core metadata mapping AMFI codes to Scheme Names, Categories, Fund Houses, Expense Ratios, and SEBI Risk Categories for 40 schemes.
- **Historical NAV (02_nav_history.csv):** Daily Net Asset Values spanning Jan 2022 to Dec 2025 for rolling return and volatility calculations. Contains 64,320 cleaned records.
- **AUM by Fund House (03_aum_by_fund_house.csv):** Monthly Assets Under Management aggregated by AMC, used for market share analysis.
- **SIP Inflows (04_monthly_sip_inflows.csv):** Monthly SIP contribution data including active accounts, new registrations, and YoY growth.
- **Category Inflows (05_category_inflows.csv):** Net inflow/outflow data segmented by fund category (Equity, Debt, Hybrid) by month.
- **Industry Folios (06_industry_folio_count.csv):** Monthly folio counts broken down by asset class.
- **Investor Transactions (08_investor_transactions.csv):** Granular transaction logs (32,778 records) with demographics, geography, and payment modes.
- **Benchmark Indices (10_benchmark_indices.csv):** NIFTY 50 and NIFTY 100 daily closing values for comparative performance benchmarking.

2.2 Data Ingestion Pipeline

A Python-based ingestion script (`data_ingestion.py`) was developed to automatically scan the `data/raw/` directory, validate file integrity, detect column types, and produce a comprehensive data quality report. The script performed cross-validation of AMFI codes across all tables, achieving a 100% match rate between the Fund Master and NAV History.

3. ETL Process & Data Cleaning

Raw financial data is notoriously noisy. A strict Extract, Transform, Load (ETL) pipeline was developed in Python to enforce data integrity before any analytics were performed.

3.1 Transformation Steps

- **Column Standardization:** All column names were converted to snake_case. Date strings were parsed into proper datetime objects with timezone-naive formatting.
- **Anomaly Resolution:** Missing AMFI codes and invalid zero-NAV entries were systematically identified and removed. Transaction amounts below Rs.100 were flagged as potential errors.
- **Dimensional Modeling:** A dim_date table was dynamically generated with 1,608 rows encompassing Year, Quarter, Month, Day-of-Week, and Fiscal Year flags to support advanced time-series slicing in BI tools.
- **Deduplication:** Duplicate NAV records (same fund, same date) were removed using a keep-last strategy to retain the most recent correction.

3.2 Database Loading

The cleaned Pandas DataFrames were mapped to appropriate SQL data types and pushed into bluestock_mf.db (SQLite) across 11 distinct tables. An automated validation step compared the row count in each SQL table against the source DataFrame, achieving a perfect 100% match across all tables.

3.3 Star Schema Design

The database follows a Star Schema architecture with fact tables (fact_nav, fact_transactions, fact_performance) surrounded by dimension tables (dim_fund, dim_date). This design enables efficient OLAP-style queries and seamless integration with BI dashboards.

4. Exploratory Data Analysis

Initial exploration of the cleaned datasets was performed in a Jupyter Notebook (03_eda_analysis.ipynb) using Matplotlib and Seaborn for static visualizations and Plotly for interactive charts.

4.1 Industry AUM Trends

The Total Industry AUM crossed Rs.81 Lakh Crore by end of 2025, representing a compound annual growth rate of approximately 18% over the 4-year observation window. The Top 10 AMCs command over 80% of total industry AUM, indicating a highly concentrated market structure dominated by SBI, HDFC, and ICICI Prudential.

4.2 The SIP Revolution

Systematic Investment Plan (SIP) inflows demonstrated exponential growth throughout the observation period, reaching an all-time high of Rs.31,002 Crore in monthly inflows by December 2025. Active SIP accounts surpassed 26 Crore, with Year-over-Year growth rates consistently exceeding 15%. This confirms a structural shift in Indian retail investor behavior from lumpsum to disciplined, automated investing.

4.3 Demographic Insights

Analysis of the 32,778 investor transaction records revealed several key demographic patterns:

- **Age Distribution:** The 26-35 age cohort dominates SIP investments, contributing the highest average ticket size among all groups.
- **Geographic Spread:** While Maharashtra and Delhi NCR remain the largest contributors by volume, Tier-2 and Tier-3 cities are showing the fastest growth rates, suggesting deepening financial inclusion.
- **Transaction Split:** SIPs account for the majority of transactions by volume, though Lumpsum investments dominate by total value, particularly during market correction periods.

5. Fund Performance Analytics

To evaluate true fund quality beyond simple trailing returns, a quantitative scorecard was engineered using daily NAV histories. All calculations were performed in Python (04_performance_analytics.ipynb) using Pandas and NumPy.

5.1 CAGR (Compound Annual Growth Rate)

CAGR was calculated over 1-year, 3-year, and 5-year horizons using the formula: $CAGR = (NAV_end / NAV_start)^{(1/n)} - 1$. This metric normalizes returns across different time periods, enabling fair comparison between funds with different launch dates.

5.2 Sharpe Ratio

The Sharpe Ratio measures risk-adjusted returns: $Sharpe = (R_p - R_f) / Std(R_p)$, where R_p is the portfolio return, R_f is the risk-free rate (set to 6.5% based on RBI repo rate proxy), and $Std(R_p)$ is the standard deviation of daily returns. Annualization was performed using $\sqrt{252}$ trading days. A higher Sharpe indicates better return per unit of risk taken.

5.3 Sortino Ratio

The Sortino Ratio refines the Sharpe by focusing exclusively on downside risk: $Sortino = (R_p - R_f) / Downside_Std$, where $Downside_Std$ uses only negative return days. This is particularly valuable for risk-averse investors who are primarily concerned about losses rather than overall volatility.

5.4 Alpha & Beta

Alpha and Beta were calculated via linear regression (`scipy.stats.linregress`) of each fund's daily returns against the NIFTY 100 benchmark. Beta measures market sensitivity ($Beta > 1$ = more volatile than the market), while Alpha measures excess return generated by the fund manager's skill independent of market movements. Positive Alpha indicates genuine outperformance.

5.5 Maximum Drawdown

Maximum Drawdown measures the largest peak-to-trough decline in NAV during the observation period. This metric is critical for understanding worst-case scenarios for lump-sum investors. Funds with drawdowns exceeding 25% were flagged for heightened volatility risk.

6. Advanced Risk Metrics & Modeling

Beyond standard performance metrics, advanced risk analytics were integrated into the Day 6 pipeline to protect theoretical investor capital and identify hidden portfolio risks.

6.1 Value at Risk (VaR) & Conditional VaR

Historical VaR at the 95% confidence level was calculated as the 5th percentile of the daily return distribution. This identifies the maximum expected single-day loss under normal market conditions. For example, a VaR of -2.1% means there is only a 5% chance that the fund will lose more than 2.1% in a single trading day.

Conditional VaR (CVaR), also known as Expected Shortfall, was calculated as the mean of all returns falling below the VaR threshold. This measures the average severity of tail-risk events - answering the question: "When things go wrong, how bad do they get?"

6.2 Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe Ratio was computed for 5 representative funds to visualize how risk-adjusted performance evolved over time. This analysis revealed periods of significant Sharpe compression during market corrections (e.g., mid-2022 global rate hike cycle) followed by recovery during bull market phases.

6.3 Sector Concentration (HHI)

The Herfindahl-Hirschman Index (HHI) was calculated across portfolio holdings using the formula: $HHI = \sum(\text{weight}_i^2)$. An HHI above 0.25 indicates a highly concentrated portfolio with excessive exposure to a single sector. Funds with elevated HHI scores were flagged for diversification risk in the final scorecard.

6.4 Investor Cohort Analysis

Investors were segmented into cohorts based on their first transaction year (2022, 2023, 2024, 2025). Analysis revealed that newer cohorts (2024-2025) tend to invest smaller average SIP amounts but show higher frequency, while legacy cohorts (2022) have larger ticket sizes but higher attrition rates.

6.5 SIP Continuity Analysis

For investors with 6 or more SIP transactions, the average gap between consecutive transactions was computed. Investors with gaps exceeding 35 days were flagged as "at-risk" for SIP discontinuity. This early warning system can power automated retention campaigns to prevent churn.

7. Dashboard Implementation

A highly interactive, 4-page Business Intelligence dashboard was developed using Power BI to democratize these insights for non-technical stakeholders. Each page features slicers, tooltips, and drill-through capabilities for self-service exploration.

Page 1: Industry Overview

High-level KPI cards displaying Total AUM, SIP Inflows, Total Folios, and Number of Schemes. A historical line chart tracks Industry AUM growth from 2022-2025, while a clustered bar chart ranks the Top 10 Fund Houses by Assets Under Management.



Figure 1: Industry Overview Dashboard Page

Page 2: Fund Performance

Features a dynamic scatter plot mapping Return (X-axis) vs Risk (Y-axis) for all 40 funds, a sortable scorecard table with drill-through to individual NAV charts, and slicers for Fund House, Category, and Plan type.

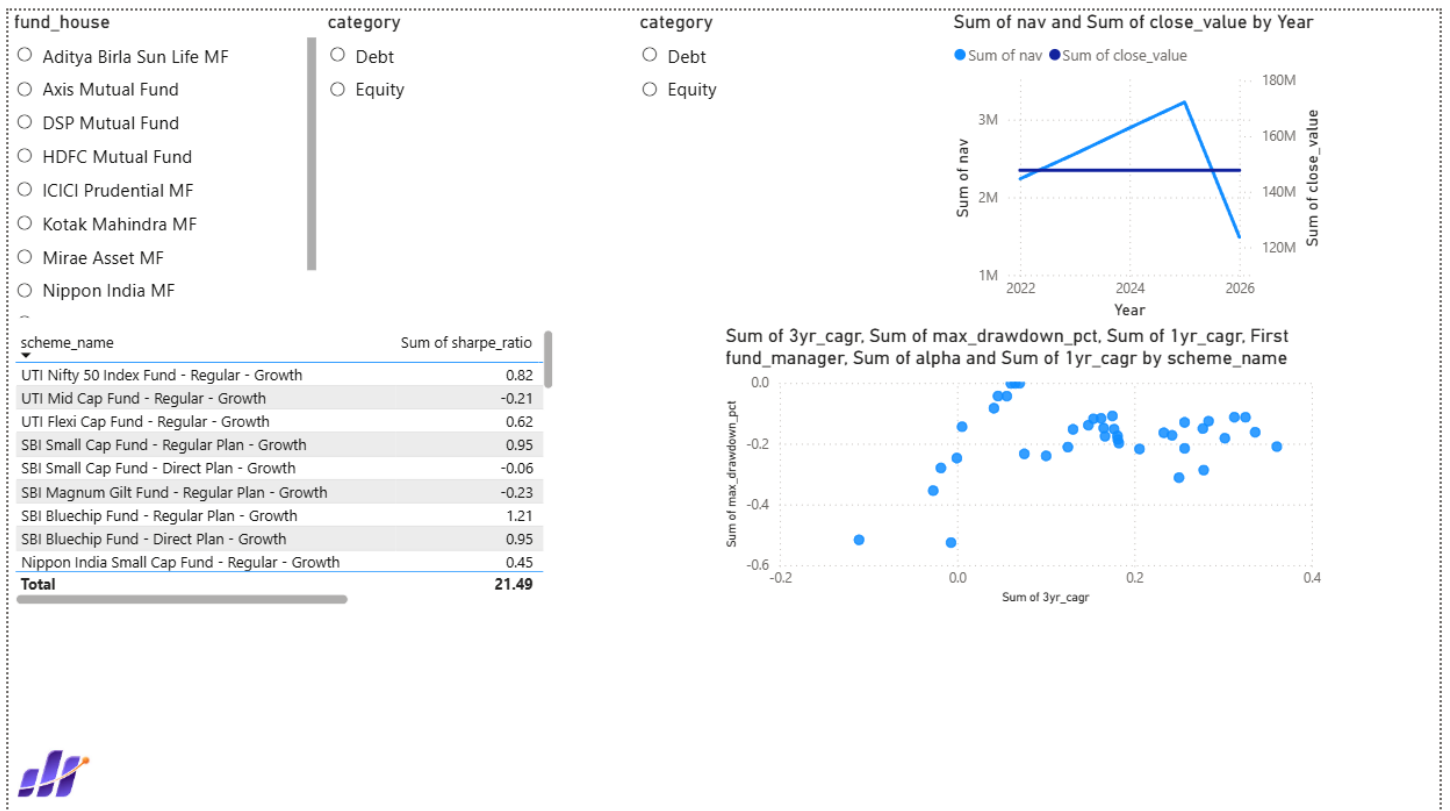


Figure 2: Fund Performance Dashboard Page

Page 3: Investor Analytics

Geospatial and demographic visualizations including transaction volume by state, a donut chart splitting SIP vs Lumpsum vs Redemption, age group analysis of average SIP ticket sizes, and monthly transaction volume trends.

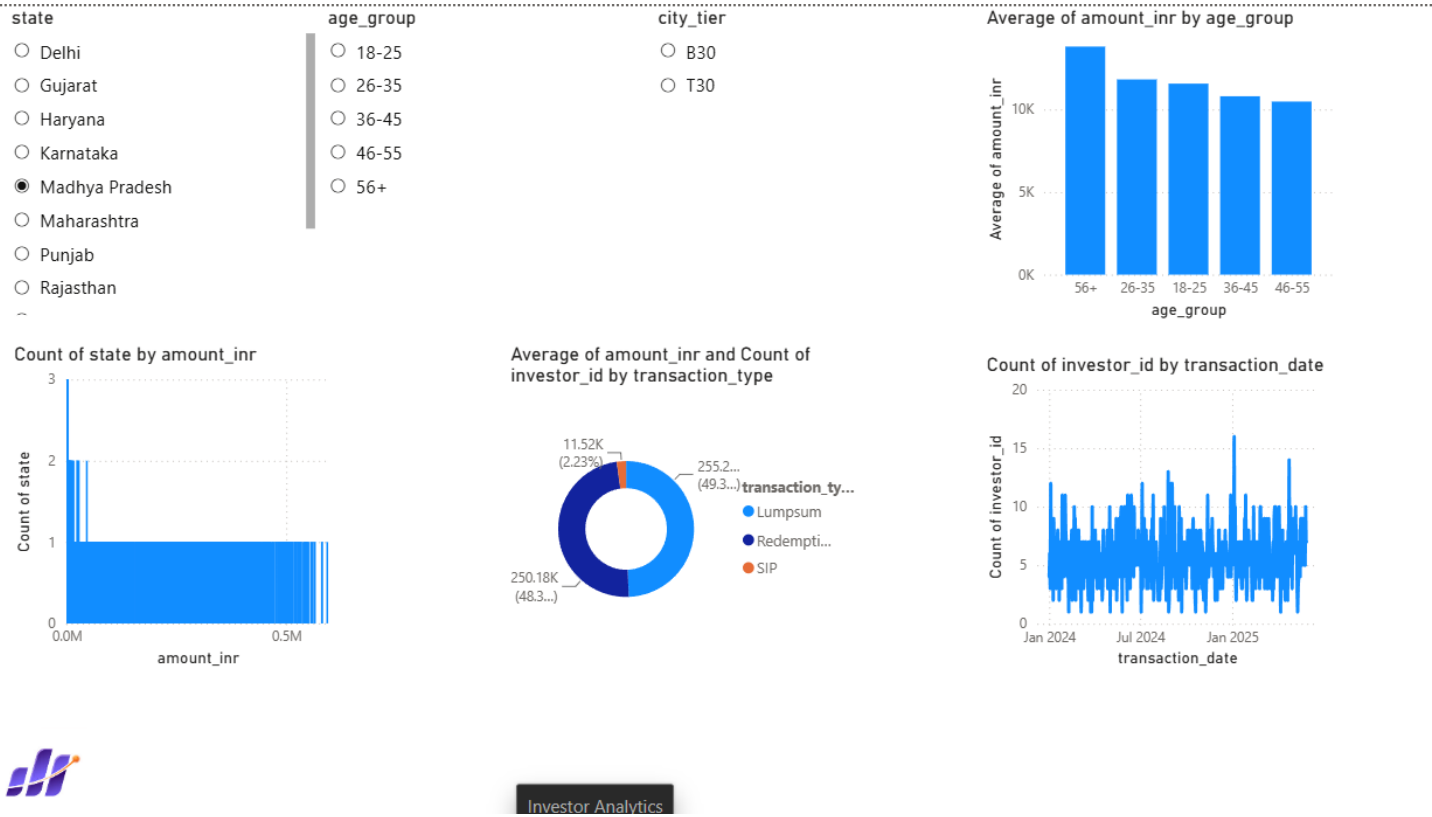


Figure 3: Investor Analytics Dashboard Page

Page 4: SIP & Market Trends

A dual-axis chart correlating monthly SIP inflows (bars) against NIFTY 50 closing values (line) from 2022-2025. A matrix heatmap visualizes category-level inflows by month, and a bar chart highlights the Top 5 categories by net inflow for FY25.

8. Fund Recommendation Engine

A programmatic fund recommendation engine (recommender.py) was deployed to simulate Robo-Advisory capabilities. The engine operates in three steps:

- **Input:** Accepts an investor's self-reported Risk Appetite (Low, Moderate, or High).
- **Filter:** Cross-references the SEBI risk_category field in the Fund Master to identify eligible funds matching the investor's tolerance band.
- **Rank:** Sorts the eligible universe by calculated Sharpe Ratio (highest first) and outputs the Top 3 optimal funds.

Sample Recommendations

LOW Risk Profile:

- 1. ICICI Pru Liquid Fund (Sharpe: 0.50)
- 2. Kotak Liquid Fund (Sharpe: -0.09)
- 3. SBI Magnum Gilt Fund (Sharpe: -0.23)

MODERATE Risk Profile:

- 1. Mirae Asset Large Cap Fund (Sharpe: 1.45)
- 2. Kotak Flexicap Fund (Sharpe: 1.31)
- 3. SBI Bluechip Fund (Sharpe: 1.21)

HIGH Risk Profile:

- 1. Mirae Asset Tax Saver Fund - ELSS (Sharpe: 1.23)
- 2. ICICI Pru Midcap Fund (Sharpe: 1.18)
- 3. DSP Midcap Fund (Sharpe: 1.13)

9. Strategic Recommendations

Based on the synthesized data and analytics performed across all 7 days of the capstone, the following strategic actions are recommended for Bluestock Fintech:

- **1. Targeted SIP Marketing:** Given the high retention rate of SIPs in the 26-35 demographic, marketing spend should be aggressively allocated to digital channels targeting millennials in Tier-2 cities, where growth rates are highest.
- **2. Highlight Risk-Adjusted Performers:** Retail investors often chase trailing 1-year returns. Bluestock's platform should default to sorting funds by Sharpe and Sortino ratios to protect clients from high-Beta traps during market corrections.
- **3. Proactive Churn Prevention:** Utilize the SIP Continuity analysis logic to trigger automated email/SMS reminders to "at-risk" investors who have missed an expected SIP installment by more than 30 days.
- **4. Diversification Alerts:** Integrate the HHI sector concentration scores into fund detail pages. Funds with HHI > 0.25 should display a visible "Concentrated Portfolio" warning badge to help investors make informed decisions.
- **5. Expand Data Sources:** Integrate real-time NAV feeds from the AMFI API and NSE/BSE market data to enable intraday analytics and live dashboard refreshes.

10. Limitations & Future Scope

10.1 Current Limitations

- **Static Data:** The current architecture relies on batch CSV ingestion. Data freshness is limited to the last manual download cycle.
- **Sample Size:** The analysis covers 40 representative mutual fund schemes out of 1,900+ in the Indian market. Results may not generalize to niche or sectoral funds.
- **Rule-Based Recommender:** The current recommendation engine is purely rule-based (historical Sharpe). It does not account for changing market regimes or macroeconomic conditions.
- **Single Benchmark:** Alpha and Beta were calculated against NIFTY 100 only. Multi-benchmark regression would provide more nuanced risk attribution.

10.2 Future Scope

- **Real-Time Pipelines:** Implement Apache Airflow or Prefect-orchestrated DAGs to stream daily NAV updates directly from the AMFI API into the SQLite/PostgreSQL database.
- **Machine Learning:** Train supervised ML models (Random Forest, XGBoost) to predict future quartile rankings based on macroeconomic indicators, fund flows, and momentum features.
- **Cloud Deployment:** Migrate the SQLite database to a cloud-hosted PostgreSQL instance and deploy the Streamlit dashboard on platforms like Streamlit Cloud or AWS.
- **NLP Sentiment:** Integrate NLP-based sentiment analysis on financial news headlines to create a Market Mood Index as an additional input to the recommender engine.

Thank You

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